

Reference: Milnor, Topology from a Differential Viewpoint & Morse Theory.

Index of a Vector Field

Recall the theorem of Sard and Brown:

Theorem (Sard): If $f : M \rightarrow \mathbb{R}^n$ is smooth, then the set of critical values has Lebesgue measure 0.

Corollary: If $f : M \rightarrow N$ is smooth, then the set of regular values is dense in N .

We can define the degree of a smooth map.

Theorem: If M, N are n -dimensional compact connected oriented smooth manifolds, and $F : M \rightarrow N$ is smooth. For $x \in M$ let $\text{sign}(x) = +1$ if dF_x is orientation preserving and $\text{sign}(x) = -1$ otherwise. Then for any regular value $y \in N$,

$$\deg(F) := \sum_{x \in F^{-1}(y)} \text{sign}(x)$$

is independent of the choice of y . Furthermore, if F, G are smoothly homotopic, then $\deg(F) = \deg(G)$.

Remark: The degree is actually defined by

$$\int_M F^* \omega = \deg(F) \int_N \omega$$

for any smooth n -form ω on N . Then taking ω as a compact cover of $F^{-1}(y)$ we obtain the result above. See GTM218 page 458.

Lemma1: If $f : M \rightarrow N$ extends to $F : X \rightarrow N$, where $M = \partial X$ is the boundary of a compact smooth manifold X with the same orientation, then $\deg(f) = 0$.

Proof: Take a n -form such that $\int_N \omega \neq 0$, then $f = F \circ \iota$ implies

$$\deg(f) \int_N \omega = \int_M f^* \omega = \int_{\partial X} F^* \omega = \int_X d(F^* \omega) = \int_X F^* d\omega = \deg(F) \int_N d\omega = 0.$$

(The third equality is Stokes' Theorem) Since the only $(n+1)$ -form on the n -dim space N is zero. Hence $\deg(f) = 0$.

Now we define the index of a vector field at its zero:

Definition: If $U \subset \mathbb{R}^n$ is open, v a vector field on U , and z an isolated zero. Take $\varepsilon > 0$ such that $B(z, \varepsilon)$ contains no other zeros, then the index $\text{ind}_v(z)$ of v at z is defined as the degree of the map

$$\varphi : C_z(\varepsilon) \rightarrow \mathbb{S}^{n-1}, x \mapsto \frac{v(x)}{\|v(x)\|}. \quad \text{where } C_z(\varepsilon) = \{x : \|x - z\| = \varepsilon\}$$

Examples: Consider electric fields.

Lemma2: Suppose $U, V \subset \mathbb{R}^n$ open, $f : U \rightarrow V$ a diffeomorphism, v a vector field on U . Let $v' = df \circ v \circ f^{-1}$ be a vector field on V , then $\text{ind}_{v'}(f(z)) = \text{ind}_v(z)$.

Lemma3: Any orientation preserving diffeomorphism of $\mathbb{R}^m$ is smoothly isotopic to the identity.

Proof of Lemma3: For any $f : \mathbb{R}^m \rightarrow \mathbb{R}^m$ WLOG assume $f(0) = 0$. Consider $F(x, t) = f(xt)/t$ for $0 < t \leq 1$ and $F(x, 0) = df_0(x)$, then since $df_0(x) = \lim_{t \rightarrow 0} f(xt)/t$, F is clearly an isotopy. To show that F is smooth as $t \rightarrow 0$, write f as $f(x) = x_1 g_1(x) + \dots + x_n g_n(x)$, then $F(x, t) = x_1 g_1(tx) + \dots + x_n g_n(tx)$ for all $t \in [0, 1]$. Hence f is isotopic to the linear map df_0 , so it is isotopic to the identity.

Proof of Lemma2: We may assume $z = f(z) = 0$ and U is convex.

First assume f is orientation preserving. By Lemma3 we have embeddings $f_t : U \rightarrow \mathbb{R}^n$ such that $f_0 = id$, $f_t(0) = 0$ and $f_1 = f$. Let $v_t = df_t \circ v \circ f_t^{-1}$ be vector fields, then there exists $\varepsilon > 0$ such that the only zero of each v_t in $B(0, \varepsilon) \subset \text{Im} f_t$ is $z = 0$. (Such ε exists: Consider a closed ball $D \subset U$ centered at 0, then $f_t(\partial D)$ is jointly continuous so the image of $\partial D \times [0, 1]$ is compact, hence the distance from 0 to the image is positive. Therefore we can take $\varepsilon > 0$ such that $B(0, \varepsilon) \subset f_t(D)$.)

Then since homotopic maps have the same degree, the Gauss maps of v_0 and v_1 have the same degree, hence $\text{ind}_{v'}(f(z)) = \text{ind}_v(z)$.

To prove the orientation reversing case, it suffices to consider when $f = \rho$ is a reflection. Then the associated gauss map $\bar{v}' = \rho \circ \bar{v} \circ \rho^{-1}$ so we can verify they have the same degree.

By this lemma, the index of vector fields on manifolds is well-defined:

Definition: If z is an isolated zero of v on M , and (U, φ) a local coordinate, then $\text{ind}_v(z)$ is defined as the index of $d\varphi \circ v \circ \varphi^{-1}$ at $\varphi(z)$.

For a vector field v , denote its index as the sum of indices of zeros of v . (as a shorthand)

Definition: If $M \subset \mathbb{R}^m$ is a compact m -dimensional manifold with boundary, then the Gauss map is defined as $g : \partial M \rightarrow \mathbb{S}^{m-1}$ which maps x to the normal unit vector, the outward vector perpendicular to $T(\partial M_x)$

Theorem: Let $M \subset \mathbb{R}^m$ be a m -dimensional manifold with boundary, v a vector field that points outward on the boundary, with no degenerate zeros. Then the index of v is the degree of the Gauss map.

Proof: Use Hausdorffness to find small spheres containing a unique zero of v . By removing these zeros we obtain another manifold N with boundary. Consider $F : N \rightarrow \mathbb{S}^{m-1}$, $x \mapsto v(x)/\|v(x)\|$, then F extends from $F|_{\partial N}$ so by Lemma1 $\deg(F|_{\partial N}) = 0$.

Since v points outward on ∂M , $F|_{\partial M}$ is homotopic to the Gauss map g : e.g. consider $W : \partial M \times [0, 1] \rightarrow \mathbb{R}^m$, $(x, t) \mapsto (1-t)\frac{v(x)}{\|v(x)\|} + tg(x)$ and the homotopy $H(x, t) = W(x, t)/\|W(x, t)\|$. Hence $\deg(F|_{\partial M}) = \deg(g)$, and by definition $\deg(F|_{C(z, \varepsilon)}) = -\text{ind}_v(z)$ for any zero z of v (since the orientation reversed when looking from the outside). Therefore $\deg(F|_{\partial N}) = 0$ implies $\sum \text{ind}_v(z) = \deg(g)$.

Corollary: Let $M \subset \mathbb{R}^k$ be a m -dimensional manifold without boundary, v a vector field on M with no degenerate zeros. For small $\varepsilon > 0$ define $N_\varepsilon = \{x \in \mathbb{R}^k : \|x - y\| \leq \varepsilon \text{ for some } y \in M\}$, then the index of v is the degree of the Gauss map on N_ε .

Proof: Consider the projection $\pi : N_\varepsilon \rightarrow M$ sending $x \in N_\varepsilon$ to its unique nearest neighbor, then π is smooth and $x - \pi(x)$ is orthogonal to $T_{\pi(x)}M$. Let $V(x) = v(\pi(x)) + (x - \pi(x))$ be a vector field on N_ε , then by orthogonality we know the zeros of V must be zeros of v .

Furthermore, suppose $v(z) = 0$, if $x \in T_z M$ then $dV(x) = dv(x)$ and if $x \in T_z M^\perp$ then $dV(x) = x$. Hence the index of V and v at z are the same.

Verify that V points outwards on ∂N_ε : $n(x) = \frac{x - \pi(x)}{\varepsilon}$ so $V(x) \cdot n(x) = (v(\pi(x)) + (x - \pi(x))) \cdot n(x) = \frac{(x - \pi(x))^2}{\varepsilon} = \varepsilon > 0$.

Since the Gauss map is independent of the vector field, we see that the index of vector field with no degenerate zeros are the same.

Applying Morse theory

For any Morse function f , denote $i_{f,j}$ as the number of critical points with index j .

Theorem: For any closed manifold M and Morse function $f : M \rightarrow \mathbb{R}$,

$$\sum_j (-1)^j i_{f,j} = \chi(M).$$

Proof: By Morse theory, $i_{f,j}$ is the number of j -cells attached when constructing the CW-complex M , so by definition it is the Euler characteristic.

As a corollary, we calculate the Euler characteristic for some manifolds:

- If M is a closed manifold of odd dimension, then $\chi(M) = 0$, since $-f$ is a Morse function, and $i_{f,j} = i_{-f, n-j}$.

- If $M = \mathbb{S}^{2n}$, then consider the height function f , with two critical points at poles. Hence $\chi(M) = (-1)^0 + (-1)^{2n} = 2$.
- If $M = \mathbb{T}$, likewise choose the height function f , then $\chi(\mathbb{T}) = (-1)^0 + (-1)^1 + (-1)^1 + (-1)^2 = 0$.

Theorem: Let M be a compact smooth manifold, there is a vector field v such that the index of v is $\chi(M)$.

Proof: Take a Morse function $f: M \rightarrow \mathbb{R}$, and let $v = \text{grad} f$. Then for each zero x of v , suppose the Morse theory index of x is j , then $\text{ind}_v(x) = (-1)^j$ since the Morse function is locally of the form $f(x) \pm x_1^2 \pm x_2^2 \pm \dots$. Therefore $\sum \text{ind}_v(x) = \sum_j (-1)^j i_{f,j} = \chi(M)$.

Poincaré-Hopf Theorem

Theorem (Poincaré-Hopf) Let v be a vector field on a compact oriented smooth manifold M (without boundary), having only isolated zeros. Then the sum of indices of the zeros of the vector field is equal to $\chi(M)$.

Proof: If all zeros of v are non-degenerate, by Morse theory we found v' such that the sum of indices of zeros of v' is $\chi(M)$, and we showed that the sum of indices of zeros of v' and v are the same. So this case was already proved.

Now suppose v has a degenerate zero at $z \in M$, and take a local coordinate $U \subset \mathbb{R}^k$ at z . Since the zeros are isolated, take $\varepsilon > 0$ such that $\overline{B}(z, 2\varepsilon) \subset U$ contain no other zero of v . Take $\delta > 0$ such that $\|v(x)\| > \delta \forall x \in \overline{B}(x, 2\varepsilon) \setminus B(x, \varepsilon)$. By Urysohn's lemma, we can take $f: U \rightarrow [0, 1]$ such that $f|_{\overline{B}(x, \varepsilon)} \equiv 1$ and $f|_{B(x, 2\varepsilon)^c} \equiv 0$. Consider the vector field $v'(x) = v(x) - f(x)y$ for $x \in U$ and $v' = v$ outside U , where y is a regular value of v such that $\|y\| < \delta$ (by Sard's theorem such y exists).

We show that v' has no degenerate zeros in U : If $x \in U \setminus \overline{B}(x, 2\varepsilon)$, then $v'(x) = v(x) \neq 0$. If $x \in \overline{B}(x, 2\varepsilon) \setminus B(x, \varepsilon)$, then $\|v(x)\| > \delta > \|y\|$ so $v'(x) = v(x) - f(x)y \neq 0$. If $x \in B(x, \varepsilon)$, then $v(x) = y$ so x is a regular point of v , hence $dv'_x = dv_x \neq 0$ so x is non-degenerate.

It suffices to show the sum of indices of zeros of z, z' coincide, then by repeatedly removing a degenerate zero, we obtain a vector field without degenerate zeros. The proof is analogous to the prior theorem.

By definition $\text{ind}_v(z)$ is the degree of $F: \partial \overline{B}(z, 2\varepsilon) \rightarrow \mathbb{S}^{n-1}, x \mapsto \frac{v(x)}{\|v(x)\|}$. Again we take small spheres in each zero z' of v' , and remove them to obtain a smooth manifold N with boundary. Likewise $\deg(F|_{\partial N}) = 0$ implies $\deg(F|_{\partial \overline{B}(0, 2\varepsilon)}) = \sum \text{ind}_{v'}(z')$ so $\text{ind}_v(z) = \sum \text{ind}_{v'}(z')$.

We have the following famous results as corollaries:

Theorem (Hairy Ball): Any smooth vector field on $\mathbb{S}^{2n}$ has at least 1 zero.

Proof: By Poincaré-Hopf the sum of indices of zeros is $\chi(\mathbb{S}^{2n}) = 2$, so the vector field cannot be non-zero.

We can easily construct a non-zero vector field on $\mathbb{S}^{2n-1} \subset \mathbb{R}^{2n}$: For $x = (x_1, \dots, x_{2n})$, let $v(x) = (-x_2, x_1, -x_4, x_3, \dots, -x_{2n}, x_{2n-1})$, then $v(x) \cdot x = 0$ and $\|v(x)\| = \|x\| = 1$.